PV3 teacher route - hundreds, tens and ones

Grade 2 • C-MATH-001 / U-MATH-001-01 • draft • range 0-999

Sequence: M01 A/B (-01), hundreds (-02), number forms (-03), counting (-04), comparison (-05), unit check (-06). Core estimate: 50 + 25 + 25 + 25 + 25 + 25 = 175 minutes, plus entry and optional work below. PVX remains separately taught.

Administration and evidence contract

Start with A-MATH-001-01 (10-15 additional minutes), using its separate learner sheet and item-specific criteria. Verify M01's twelve-counter prerequisite. State the route from actual entry and lesson evidence, not grade alone. No learner result or actual pace is assumed. Use the pack materials checklist; establish the amount represented by each grouped object before using it as a token.

The sequence uses M01 two-cycle communication, then worked-example fading in number forms, retrieval and concrete boundary modeling in counting, and a claim/counterexample investigation in comparison. Complete models come before supported attempts and blank independent sheets. Fading removes solution cues, not manipulatives, read-aloud, scribing or explanation criteria. A teacher can be the partner; no compulsory online practice or peer availability is assumed.

For lessons 2-5 use 3 minutes retrieval, 5 modeling, 7 guided practice, 7 independent work and 3 exit/check. Split or extend when needed. Independent sheets revisit some guided claims for practice; these are formative, not fresh summative evidence. Give substantive help when needed and recognize the resulting work as supported. All prompt parts have keys below or in the explicitly linked M01/assessment key. These are authoring estimates, not a calendar or a required performance speed.

Lesson 1 - L-MATH-001-01-01 using M01

Outcomes: O-MATH-001-001 tens/ones component; preparatory 002-003. Prerequisite: count twelve counters one-to-one, name one counter as one-unit. Materials: enough counters/base-ten equivalents for 99, blank Tens/Ones chart, separate partner cards, paper and pencil. Primary art: arithmetic grouping.

Use the merged M01 teacher guide with its unchanged learner book, workbook and separate cards. Its scripts and complete A1-B4, W1-W8/card keys are authoritative

and included in the teacher print edition. Cycle A models 14, reconstructs groups, and ends with 27; Cycle B unpacks 21 and ends with two models of 46. Grammar names the counted one-unit and grouped ten; logic conserves the collection; rhetoric asks a partner to reconstruct and inspect the explanation.

Use A4/B4 criteria: accurate amount and labels, conservation reason and explanation another person can follow. Failure to count requires smaller collection instruction; unit confusion requires opening a ten; zero confusion requires contrasting grouped tens with an empty collection. Continue only after evidence in this component. Do not read this as whole outcome-001 success. The optional A-MATH-001-04 review has four tasks and a separate teacher key, linked in the pack and included in print. Core time is 25 + 25 = 50; workbook 10-16 and formative review 5-10 are additional.

Lesson 2 - L-MATH-001-01-02: hundreds and zero positions

Outcomes: O-MATH-001-001. Prerequisite: M01 conserving ten/ones exchange and zero ones explanation. Materials: established ten-groups, hundred-groups with inspectable contents, enough base-ten equivalents for 999, blank H/T/O chart, paper, pencil, learner sheet. Primary art: arithmetic decimal grouping.

Retrieve 40. Count ten tens as 10, 20, ... , 100 and exchange those tens for one hundred; reopen the hundred to show the same amount. Compare 100 and 200. Guided work: build 140, 305, 350; in 140 unpack one hundred into ten tens, then one ten into ten ones. Keep the entire collection at each stage; never count a bundle and its contents twice. Use sheet 01-03 for independent work; 04 is the exit. Grammar distinguishes digit and place amount; logic tests the same-digits claim and conserves exchanges; rhetoric explains zero tens to a partner.

Complete lesson and learner key

Retrieval: 40 = 4 tens. Ten tens = 100 ones = 1 hundred; 200 = 2 hundreds. Guided: 140 = 1 hundred + 4 tens + 0 ones; 305 = 3 hundreds + 0 tens + 5 ones; 350 = 3 hundreds + 5 tens + 0 ones.

- P-PV3-02-01: 406 = 4 hundreds + 0 tens + 6 ones; zero tens means no extra tens. 230 = 2 hundreds + 3 tens + 0 ones; zero ones means no extra singles. Neither zero means the whole collection is empty.

- P-PV3-02-02: 140 = 1 hundred + 4 tens + 0 ones = 14 tens + 0 ones = 13 tens + 10 ones. Each exchange substitutes equal-valued groups; all 140 ones remain.

- P-PV3-02-03: false; 305 = 300 + 5, whereas 350 = 300 + 50. The 5 contributes five ones or five tens respectively, so the amounts differ.

- P-PV3-02-04: 402 = 4 hundreds + 0 tens + 2 ones; zero records no extra tens. Ten tens and one hundred both contain 100 ones, so exchange changes groups without changing amount.

Success check: correct models and zero explanations for all asked positions, every reverse exchange shown, and ten tens as a conserved hundred. Return to open groups if a token is named without its amount; do not advance based only on writing a three-digit numeral. This does not teach or certify a thousand.

Lesson 3 - L-MATH-001-01-03: number forms

Outcomes: O-MATH-001-003; review 001. Prerequisite: lesson 2's hundred/tens exchange and zero positions. Materials: H/T/O models, blank chart, paper, pencil and learner sheet. Primary art: arithmetic representation.

Retrieve 305. Model 257, saying "two hundred fifty-seven," and connect 200 + 50 + 7 to the labeled groups. Guided work: 340 and 608. Test 608 = 60 + 8 by building both amounts before revealing the correction. Remove the completed model for independent sheet items 01/03/04; item 02 is the exit. Grammar names the forms; logic tests a place-unit mismatch; rhetoric explains agreement among name, numeral, sum and model.

Complete lesson and learner key

Retrieval: 305 = 3 hundreds + 0 tens + 5 ones, three hundred five. Model: 257 = 200 + 50 + 7; 2 hundreds + 5 tens + 7 ones. Guided: 340, three hundred forty, 300 + 40; 608, six hundred eight, 600 + 8.

- P-PV3-03-01: 503 = 500 + 3; five hundred three; model 5 hundreds, 0 tens, 3 ones.

- P-PV3-03-02: four hundred sixty; 400 + 60; the 6 contributes six tens, 60; zero means no extra ones.

- P-PV3-03-03: false; 60 + 8 = 68, whereas 608 = 600 + 8. The 6 in 608 records hundreds, not tens.

- P-PV3-03-04: three hundred forty; 300 + 40; 3 hundreds and 4 tens give the same 340 one-units in all forms.

Success check: all requested representations agree, each digit's contribution is named accurately, and the zero/unit reasoning is clear. Spoken names and spelling support are allowed; optional zero sum terms do not change correctness. Return to

the physical groups on a mismatch. The optional thousand activity is outside this core.

Lesson 4 - L-MATH-001-01-04: steps and hundreds boundary

Outcomes: O-MATH-001-002; review 001/003. Prerequisite: read the amounts and explain both hundred/tens and ten/ones exchanges. Materials: groups, blank chart, paper, pencil and learner sheet. Primary art: arithmetic repeated addition.

Retrieve 98 through 102 with models. At 99 add one: 9 tens + 10 ones; exchange to 10 tens, then to one hundred. Model four terms by tens from 230. Guided work: hundreds from 500 to 900, naming one hundred as the step. Use sheet 01/04 independently; 02/03 for exit and explanation. Grammar defines start, term and step; logic separates addition from conserving exchanges; rhetoric teaches the rule with a chart. Starting values count as first terms, endpoints are included.

Complete lesson and learner key

Retrieval: 98, 99, 100, 101, 102. Model: 230, 240, 250, 260; add one ten each time. Guided: 500, 600, 700, 800, 900; add one hundred each time.

- P-PV3-04-01: 95, 100, 105, 110, 115; step five.

- P-PV3-04-02: 198, 199, 200, 201, 202. Before: 1 hundred + 9 tens + 9 ones. Add one to obtain 1 hundred + 9 tens + 10 ones; exchange to 1 hundred + 10 tens + 0 ones; then to 2 hundreds + 0 tens + 0 ones. Exchanges conserve the new 200 after addition.

- P-PV3-04-03: 390, 400, 410. Adding one ten to 3 hundreds + 9 tens gives 3 hundreds + 10 tens; exchange to 4 hundreds. Ones remain zero.

- P-PV3-04-04: 500, 600, 700, 800, 900; each step adds one hundred, equivalent to ten tens or one hundred ones.

Success check: all terms/steps correct with requested models, every exchange and a preservation reason. A recited list alone is insufficient. Rebuild a shorter boundary span if needed before progression. The lesson's thousand boundary activity requires separate teaching and is excluded here.

Lesson 5 - L-MATH-001-01-05: comparison

Outcomes: O-MATH-001-004; review 001/003. Prerequisite: connect three-digit numerals to grouped amounts. Materials: H/T/O models, blank chart, paper, pencil and learner sheet. Primary art: arithmetic order.

Retrieve 204 and 240. Model 415 versus 451: match hundreds, then compare tens. Guided work: 390 versus 409 and 600 versus 600. Test the irrelevant-digit argument with groups. Use sheet 01/03/04 independently; 02 is the exit. Grammar reads complete relation statements; logic explains why the first differing place controls order; rhetoric states claim, model and reason for a recipient.

Complete lesson and learner key

Retrieval: 204 = 2 hundreds + 0 tens + 4 ones; 240 = 2 hundreds + 4 tens + 0 ones. Model: 415 < 451; hundreds agree, one ten is less than five tens. Guided: 390 < 409; three hundreds are less than four; 600 = 600.

- P-PV3-05-01: 204 < 240 < 420. First pair: hundreds agree and zero tens are less than four tens. Second: two hundreds are less than four hundreds.

- P-PV3-05-02: 708 < 780; hundreds agree, zero tens are less than eight tens. 659 > 599; six hundreds exceed five hundreds.

- P-PV3-05-03: claim false; 390 < 409. Hundreds differ first; the larger tens digit of 390 cannot outweigh the extra hundred in 409.

- P-PV3-05-04: 600 = 600; same six hundreds, zero tens and zero ones in both models.

Success check: symbols, complete relations and first-differing-place/equality reasons agree for every requested comparison. Restore matched place-value models on irrelevant-digit reasoning. Use PV3 unit evidence only after the route's relationships have been taught, without requiring the optional PVX extension.

Lesson 6 - L-MATH-001-01-06: selected-route evidence

Outcomes: O-MATH-001-001-004 within the taught route. Materials: selected learner form, working paper, counters/base-ten blocks, blank chart; no calculator. Prerequisite: the route's preceding lesson models and explanations, or equivalent privately documented evidence. Do not assess PVX without its separate instruction.

Retrieve unit names and symbols for 3 minutes without exposing assessment answers. Distribute only the selected A-MATH-001-02 learner form. Allow 15-20 minutes, extra time or split sittings. Complete a 2-minute teach-back if useful, but

inspect all requested reasons during the assessment, not only that final sample. No speed threshold applies. Teacher print editions include the full existing assessment key; do not distribute those editions to learners.

Read directions without answer cues; accept speech, drawing, pointing and scribing. Accept customary "and" in number names and omit zero terms in sums if accurate. Record substantive teaching prompts as support. Every subpart must meet its outcome criterion: model/units/exchange/zero in items 01-03 plus composition in 04; all sequences, steps and boundary reasons in 04-06; conversions and model connection in 07-09; comparison symbols and place-value reasons in 10-12. A recited sequence alone does not establish the conservation criterion in item 04. Addition changes the amount; subsequent equal-valued exchanges conserve the new amount.

Grammar checks names and notation; logic checks invariant/step/comparison reasons; rhetoric checks whether the explanation connects claim, model and reason for another person. Inspect independently / supported / needs review / not assessed per outcome and taught scope privately. Do not sum these into a course pass mark.

Use a fresh inspected bank 1.0.1 conceptual form when full applications were practiced or keys exposed. Changing seeds alone does not establish freshness. Keep manifests and all keys separate. Timed bank drills remain optional fluency work outside the core instruction/assessment budget, and never replace model/reason evidence.

Follow-up decisions by relationship

Evidence gap	Revisit	Fresh application / next decision
One-unit, grouped unit or reverse exchange (001)	M01; PV3 also lesson 2	Open and repack equal-valued groups; show each stage with a different amount before continuing
Zero position (001/003)	M01 zero ones; PV3 lesson 2 zero tens	Contrast an empty position with an empty collection; name remaining groups
Boundary or repeated step (001/002)	Lesson 4 objects and chart	Add the one or grouped step first, then conserve that new amount through exchanges; check a different boundary
Name, numeral or sum mismatch (003)	Lesson 3 concrete-to-symbol connection	Use a new keyed amount and connect each place to its unit; spelling support remains available
Irrelevant digit used for comparison (004)	Lesson 5 matched higher places	Build two fresh keyed amounts; inspect the first differing place or equality
Correct result without model/reason	Relevant lesson explanation task	Ask for a demonstration; recognize teaching prompts as support and recheck afterward

Proceed in the checked scope only after every part of the relevant outcome group is accurate with its model and reason. Supported evidence informs teaching; obtain fresh independent application before calling that relationship independent. Keep unassessed content pending. No duration, confidence rating or exercise count proves understanding. Private records include route/form ID, support, outcome judgments and follow-up; the shared repository receives only curriculum changes.

Optional PVX after PV3

Use the original lesson-3 thousand activity and lesson-4 boundary activity linked from the pack index, with their actual independent practice and complete keys. Teacher print editions append those unchanged lesson records for reference. Prerequisite: model 200 and explain the hundred-to-ten relationship, then demonstrate thousand grouping and all three boundary exchanges before administering PVX. Materials: ten established hundreds, a thousand-group, blank four-column chart. Additional instruction is 25 + 10-15 minutes; assessment adds 5-10, with flexibility. Assess only grouping/representation of 1,000 and the 996-1,000 boundary, not general four-digit calculations or comparison. PVX-01 and -02 have their existing full key in A-MATH-001-02. Omit entirely when untaught; record pending rather than failed.

M01 – Teacher guide and complete key

- Implements: L-MATH-001-01-01

- Learners: 2nd grade; PV2 scope, 0-99

- Status: draft; readiness review pending

- Evidence: O-MATH-001-001, tens/ones component only

- Materials: at least 99 counters or base-ten equivalents, ten-bundles/chart, paper, pencil, separate learner/workbook copies and one partner card each

- Access: read directions aloud; allow objects, drawing, pointing, spoken explanations and an adult scribe; no reading-speed or spelling criterion

Use the existing entry check and count twelve counters one-to-one before starting. Confirm the same kind of counter is the one-unit throughout an exchange. If counting is insecure, return to a smaller collection; do not silently rewrite the fixed pilot and call it reviewed.

Cycle A - Build and describe

Stage	Action	Planned minutes
Entry/retrieval	Count twelve and name the one-unit	3
Model	Group fourteen counters into one ten and four ones; show all fourteen are retained	5
Guided work	A1 and A2; have the learner point to the counted units	7
Partner exchange	A3 using separately distributed cards; swap describing/reconstructing roles	7
Exit	A4, with a model and conservation reason	3

Original modeled exchange: Teacher: “These fourteen counters are our amount. I put ten in one group. What stayed the same?” Partner: “All fourteen counters are still here.” Teacher: “What do we call that group?” Partner: “One ten.” Ask the learner to

demonstrate why the statements are true, rather than memorizing the dialogue.

A2 script, read one collection at a time without showing this guide: “Build three tens and four extra ones.” Then: “Build six tens with no extra ones.” The read-aloud description is the input, not an answer cue. Additional help in interpreting a unit or choosing a group is support and should be recognized as such.

Cycle B - Exchange and explain

Stage	Action	Planned minutes
Retrieval	Rebuild A4 and name the grouped units	3
Model	Start with 21 as 2 tens + 1 one; unpack one ten into 10 ones, leaving 1 ten + 11 ones	5
Guided work	B1 and B2; test the false claim physically	7
Partner exchange	B3 with the second section of the separate cards	7
Exit	B4; seek the amount-preservation explanation	3

An unpacked ten becomes ten individual ones. Do not remove a ten without adding its ten ones, count a bundle as a single one, or add a ten and its contents to the same model. Packing is the reverse change. Make all stages inspectable.

Learner-book and card key

Task	Exact result/model	Required reason or observation
Opening model	14 = 1 ten + 4 ones	Ten of the original one-units make the grouped ten
A1	16 = 1 ten + 6 ones	Grouping retains all sixteen counted units
A2, first	34 = 3 tens + 4 ones	Correct reconstruction from named units
A2, second	60 = 6 tens + 0 ones	Zero means no extra single ones beyond the tens
A3, red/card A	24 = 2 tens + 4 ones	Partner's model agrees with the stated groups
A3, blue/card B	50 = 5 tens + 0 ones	No extra ones; fifty ones are contained in the five tens
A4	27 = 2 tens + 7 ones	Identify a single counter as the one-unit; grouping preserves 27
B1	32 = 3 tens + 2 ones = 2 tens + 12 ones	One ten is replaced by ten ones
B2	Claim false: 43 = 4 tens + 3 ones = 3 tens + 13 ones	The proposed 3 tens + 3 ones gives 33 and loses ten
B3, green/card A	1 ten + 16 ones = 2 tens + 6 ones = 26	Pack ten of the sixteen ones; all 26 remain
B3, gold/card B	3 tens + 14 ones = 4 tens + 4 ones = 44	Pack ten of the fourteen ones; all 44 remain
B4	46 = 4 tens + 6 ones = 3 tens + 16 ones	The same forty-six one-units are represented

A partner's agreement is feedback, not proof of correctness. Check the actual reconstruction and named units. For a single learner, the teacher holds the other card and models the recipient role; no peer access is required.

Workbook key

Task	Exact result/model	Evidence
W1	28 = 2 tens + 8 ones	Groups and labels agree
W2	4 tens + 9 ones = 49	Numeral agrees with the model
W3	70 = 7 tens + 0 ones	Zero records no extra ones, not an empty collection
W4	51 = 5 tens + 1 one = 4 tens + 11 ones	One ten becomes ten ones
W5	2 tens + 15 ones = 3 tens + 5 ones = 35	Ten of the fifteen ones form another ten
W6	Ten ones equal one ten	One one would have value 1, not 10
W7	38 = 3 tens + 8 ones = 2 tens + 18 ones	Exchange preserves 38
W8	W5 represents 35 throughout	One group of ten replaces ten separate units; none enter or leave

Accept equivalent accurate models and spoken explanations; correct a unit name when it makes the mathematical meaning wrong. No particular sentence form is required. These tasks rehearse components of outcome 001 and do not certify outcomes 002-004.

Evidence and next step

A4/B4 exits: require correct unit labels, an accurate model and a conservation reason the recipient can follow. If the answer is correct but the reason is missing, ask for a demonstration and treat any teaching prompt as support. For unit/amount confusion, return to the physical one-unit and a ten-group. For zero confusion, contrast a collection of tens with an empty collection. For an exchange error, preserve the full collection and reverse the exchange.

Core sessions total 50 planned minutes and replace the usual 25-minute lesson. Optional workbook blocks total another 10-16 minutes; the separate formative review plans another 5-10, with flexibility. Review time is not a speed threshold. Use the outcome judgment to plan review or the selected next lesson; the PV3 route still

needs lesson 2. The pilot alone does not establish hundreds, counting sequences, number-form conversions or comparison outcomes.

Keep the guide and review key out of learner view. Published/practiced tasks require fresh keyed applications before independent assessment claims. Individual learner evidence remains private.

Second-grade mathematics entry check

- Assessment ID: A-MATH-001-01
- Course: C-MATH-001
- Intended grades: 2nd grade
- Purpose: prerequisite check and placement conversation
- Status: draft
- Estimated time: 10-15 minutes; no speed requirement
- Materials: counters, paper, pencil, optional tens/ones chart
- Support: read directions aloud; allow objects, drawings, and spoken responses
- Authoring review: 2026-10-09; first-unit review
- Readiness review: pending

Prompts and teacher key

These are original school-ops prompts, not retrieved Khan Academy questions.

Item	Prompt	Teacher key and evidence
1	Count a set of 18 counters, then show a different arrangement with the same total.	18; each object counted once, total remains 18
2	Count aloud from 96 through 102.	96, 97, 98, 99, 100, 101, 102
3	Write the numeral for sixty-three; say how many tens and ones it has.	63; 6 tens and 3 ones
4	Which is greater, 54 or 45? Explain with objects or a drawing.	54; 5 tens exceed 4 tens
5	Find 8 + 7 and describe a method.	15; e.g. 8 + 2 + 5

Item	Prompt	Teacher key and evidence
6	Find 14 - 6 and show what the subtraction means.	8; appropriate removal, difference, or missing-addend model

Use the evidence

Record each skill privately as independent, demonstrated with support, or needing review. There is no aggregate pass mark and no assumed learner result.

- Counting/notation gaps: add short object-counting and numeral practice before moving beyond the first place-value lesson.

- Tens/ones gaps: use lesson 1 concretely and revisit exchanges before hundreds.

- Arithmetic gaps: plan the within-20 fluency strand alongside place-value work.

- Confident explanations: consider shortening review after checking fresh examples.

This check samples entry skills. It does not certify prior-grade coverage or course mastery, and it does not assess the full second-grade outcome set.

Standalone administration and item criteria

Distribute only entry-tasks.md, working paper and a blank chart. Keep this guide out of view. Plan 10-15 minutes with extra time or split sittings; no calculator, deadline or aggregate pass mark. Read directions without solution cues; allow objects, speech, drawing, pointing and adult scribing. Recognize substantive teaching prompts as support, then give feedback after the attempt. The learner E1-E6 labels correspond to items 1-6 above, preserving this assessment ID.

Entry skill / item	Sufficient evidence for the sampled skill	Follow-up
One-to-one counting / E1	Exactly 18, each object once; rearrangement retains all objects and total	Smaller collections if uncertain; also check M01's count of twelve
Boundary count / E2	96, 97, 98, 99, 100, 101, 102 in order	Diagnostic only; difficulty beyond 99 does not block PV2
Tens/ones notation / E3	63 with six tens and three ones, matching grouped units	Use M01 concretely for gaps

Entry skill / item	Sufficient evidence for the sampled skill	Follow-up
Comparison / E4	54 > 45, with groups or drawing showing five tens exceed four	Revisit modeled amounts; lesson 5 later teaches the relationship
Composition / E5	15 and a valid method, such as making ten from 8 + 2 then adding 5	Parallel within-20 practice, not a place-value gate
Difference / E6	8 and a valid removal, difference or missing-addend model	Parallel arithmetic instruction; not a blanket place-value exclusion

Record independent / supported / needs review / not assessed by skill privately. This sampling does not establish PV3 prerequisites: require the lesson-1 and lesson-2 exchanges and zero-position explanations before the larger route. See the 2026-10-27 audit. Status remains draft; no actual entry evidence has been recorded.

Place-value unit check

- Assessment ID: A-MATH-001-02

- Course/unit: C-MATH-001 / U-MATH-001-01

- Intended grades: 2nd grade

- Purpose: end-of-unit evidence for outcomes 001-004 in the taught scope

- Status: draft

- Estimated time: 15-20 minutes for a core form; allow extra time or separate sittings

- Extension estimate: another 5-10 minutes when the 1,000 work has been taught

- Materials: paper, pencil, counters or base-ten blocks, place-value chart; no calculator

- Support: directions read aloud; objects, spoken/drawn explanations allowed; no answer cues

- Authoring review: 2026-10-09; first-unit review

- Readiness review: pending

Choose the taught range

Choose a form from the work actually taught and demonstrated in entry and lesson checks. Grade level alone does not choose the form. Record the form, taught range, and permitted support privately before administering it.

Form	Learner task sheet	Content checked	Content still unassessed
PV2	Two-digit tasks	Ones/tens, zero ones, two-digit representations and comparisons, repeated steps within 99	Hundreds and the 1,000 boundary

Form	Learner task sheet	Content checked	Content still unassessed
PV3	Three-digit tasks	Ones/tens/hundreds, zero positions, three-digit representations and comparisons, repeated steps within 999	The 1,000 boundary
PVX	Optional 1,000 extension	Ten hundreds as one thousand; four-digit notation and counting through 1,000	Other four-digit calculations or comparisons

PVX supplements PV3 only after grouping ten hundreds, the thousand position, and counting through that boundary have been taught: use lesson 3's optional thousand activity, then lesson 4's boundary activity. Their instruction time is additional to the extension assessment estimate. PVX is not a prerequisite for two-digit or three-digit progression. PV2 evidence supports only the two-digit portion of the course outcomes; it does not establish the hundreds component of outcome 001. Record unassessed extensions as pending, not as failures or completed outcomes.

Each core form has its own tasks, key, and the criteria below. If a still smaller range or a different task is needed, prepare its actual task, key, and corresponding criteria for review before use; do not treat an improvised change as this form.

Administration

The linked task sheets contain original school-ops questions without teacher keys or links to this answer guide. Their non-clickable assessment ID preserves the connection to this teacher record. Distribute a standalone copy or export of only the chosen learner sheet and any blank chart; keep this guide and lesson keys out of view during the check. Repository navigation exposes teacher material, so use the learner copy for administration. Read directions without explaining the answer. Where a number of counting terms is requested, the starting value is the first term; range endpoints are included.

The forms use different application examples from the lesson practice, including the former 95-by-fives exercise. Basic unit relationships are intentionally taught; the checks ask for models/reasons and their application to new amounts. If a full assessment prompt has already been practiced or its key seen, prepare a fresh keyed application for the same relationship before seeking independent evidence.

Merely hiding a previously seen key does not make the task fresh.

Allow objects, drawings, and spoken explanations. Accept accurate spoken number names when spelling needs support; customary use of "and" in a number name does not change its value. Do not impose a speed threshold. Give feedback after the attempt. Record answer cues or teaching prompts as support, rather than calling the resulting response independent.

Two-digit teacher key

Use with PV2 only. Zero terms in expanded form are optional.

Item	Teacher key or model evidence
PV2-01	47 = 4 tens + 7 ones = 40 + 7; identify the single object as the one-unit and each ten-group as ten such units
PV2-02	34 = 3 tens + 4 ones = 2 tens + 14 ones; one ten becomes ten ones, so the total amount remains 34
PV2-03	50 = 5 tens + 0 ones; 80 = 8 tens + 0 ones; zero means no extra single ones beyond the grouped tens
PV2-04	67, 68, 69, 70, 71; at 69 to 70, ten ones become one ten
PV2-05	15, 20, 25, 30, 35; the step is five, including 15 as the first term
PV2-06	23, 33, 43, 53, 63; the step is ten; each term gains one ten while the three ones remain
PV2-07	Seventy-two; 70 + 2; model 7 tens + 2 ones and connect the written places to those amounts
PV2-08	86; 80 + 6
PV2-09	30; thirty; 30 (or 30 + 0); the zero records no additional ones
PV2-10	64 > 46; six tens exceed four tens
PV2-11	38 < 42; three tens are less than four tens
PV2-12	55 = 55; both represent the same five tens and five ones

Three-digit teacher key

Use with PV3 only. Zero terms in expanded form are optional.

Item	Teacher key or model evidence
PV3-01	374 = 3 hundreds + 7 tens + 4 ones = 300 + 70 + 4; identify the one-unit and the ten/hundred groups made from it
PV3-02	264 = 2 hundreds + 6 tens + 4 ones = 1 hundred + 16 tens + 4 ones = 1 hundred + 15 tens + 14 ones; each exchange preserves 264
PV3-03	507 = 5 hundreds + 0 tens + 7 ones; 300 = 3 hundreds + 0 tens + 0 ones; explain each zero position
PV3-04	397, 398, 399, 400, 401; at 399 to 400, exchanging ten ones and then ten tens supplies another hundred
PV3-05	45, 50, 55, 60, 65; the step is five, including 45 as the first term
PV3-06	Tens: 280, 290, 300, 310, 320; hundreds: 200, 300, 400, 500, 600; identify ten and one hundred as the respective steps
PV3-07	Seven hundred nine; 700 + 9; model 7 hundreds + 0 tens + 9 ones and connect the written places to those amounts
PV3-08	630; 600 + 30
PV3-09	380; three hundred eighty; 300 + 80 (or 300 + 80 + 0)
PV3-10	648 < 684; hundreds agree, then four tens are less than eight tens
PV3-11	503 > 480; five hundreds exceed four hundreds
PV3-12	720 = 720; both represent the same seven hundreds, two tens, and zero ones

Optional 1,000 teacher key

Item	Teacher key or model evidence
PVX-01	Each hundred represents 100 ones, so ten hundreds represent 10 × 100 = 1,000 ones. Exchanging those groups for one thousand preserves that amount; no ones are added or removed. Numeral: 1,000; name: one thousand; place-value sum: 1,000 (or 1,000 + 0 + 0 + 0). The model/chart has one thousand, zero hundreds, zero tens, and zero ones; four digit positions, with the comma only a separator
PVX-02	996, 997, 998, 999, 1,000; at 999 to 1,000, ten ones become one ten, ten tens become one hundred, and ten hundreds become one thousand

1,000 is a four-digit numeral. PV3 comparison tasks use three-digit values only; the optional extension does not claim four-digit comparison coverage.

Outcome criteria

These are proposed local teaching criteria, applied separately to the selected form and taught scope. Check every part of a multi-part item. A correct answer alone is insufficient when the task also asks for a reason or model.

Course outcome	Core items	Proposed evidence to progress in the checked scope
O-MATH-001-001	PV2/PV3 items 01-03 and the boundary composition in item 04	All models and digit amounts correct; name the counted unit, explain unpacking and each zero position. In item 04, demonstrate composing ten ones into one ten (PV2), or ten ones into one ten and then ten tens into one hundred (PV3), preserving the amount after adding the next one
O-MATH-001-002	PV2/PV3 items 04-06	Complete sequences correct, including endpoints/starting terms; identify each step and explain the boundary exchange in item 04
O-MATH-001-003	PV2/PV3 items 07-09	All numeral, name, and place-value conversions correct; connect item 07 to its model
O-MATH-001-004	PV2/PV3 items 10-12	All comparisons and symbols correct; justify the first differing place or equality

Optional extension coverage	Items	Proposed evidence
O-MATH-001-001: thousand grouping	PVX-01 and the exchange explanation in PVX-02	Explain preservation of the amount when ten hundreds become one thousand and the resulting zero lower positions
O-MATH-001-002: count through 1,000	PVX-02	All five terms correct; explain the boundary exchange

Optional extension coverage	Items	Proposed evidence
O-MATH-001-003: thousand representation	PVX-01	Correct numeral, name, place-value sum, and four-position model

Across each group, inspect grammar (names and representations), logic (the reason), and rhetoric (an explanation another person can follow). Keep judgments separate by relationship and form: independent, demonstrated with support, needing review, or not assessed. Do not replace these judgments with one total score.

Item 04 supplies composition evidence for outcome 001 as well as counting evidence for outcome 002. Check both parts: a correct recited sequence alone does not demonstrate composition, and a correct exchange alone does not complete the count.

When evidence is incomplete, revisit the relevant lesson and reassess with fresh keyed examples in the taught scope. Larger-range work remains pending until taught and checked. Keep individual evidence privately and use curriculum observations to revise pacing.

Tens and ones – formative cycle review

- Assessment ID: A-MATH-001-04

- Course/unit: C-MATH-001 / U-MATH-001-01

- Lesson/module: L-MATH-001-01-01 / M01 pilot

- Intended learners: 2nd grade, taught tens/ones range 0-99

- Status: draft

- Purpose: formative check of the tens/ones component of O-MATH-001-001

- Time estimate: 5-10 minutes; permit extra time or split sittings, no speed threshold

- Materials: paper, pencil, counters/base-ten blocks or blank place-value chart; no calculator

- Permitted support: directions read aloud, spoken/drawn/pointed responses, adult scribe

- Readiness review: pending

Administration and scope

Distribute only the learner task sheet. No model answers, worked examples or answer cues during the attempt; discuss feedback afterward. State the taught scope and support used privately. If a full task or key has been exposed, use a fresh reviewed keyed application before calling the response independent. A public repository is not a secure assessment delivery interface.

This is formative cycle evidence, not the whole unit assessment. It does not replace A-MATH-001-02/PV2, certify hundreds or establish outcomes 002-004. Count the final amount after the addition in R4; conservation applies to the exchange, not to the preceding addition.

Key and outcome-specific criteria

Every task checks a component of O-MATH-001-001.

Task	Key/model evidence	Criterion
R1	62 = 6 tens + 2 ones; one ten contains 10 of the named single counters	Correct amount, units and ten-to-one-unit relationship
R2	48 = 4 tens + 8 ones = 3 tens + 18 ones	Both models correct; the ten is replaced by ten ones without removing amount
R3	80 = 8 tens + 0 ones	Zero means no extra single ones; eight tens still contain eighty one-units
R4	19 + 1 gives 1 ten + 10 ones = 2 tens + 0 ones = 20	Show both stages after addition; packing preserves the twenty one-units

Judge each task as independent, supported, needs review or not assessed. Within this checked component, seek all four accurate models plus the requested unit, zero and conservation reasons. An accurate numeral alone is insufficient. A model/explanation may be spoken, drawn or demonstrated without a fixed verbal formula. Report the tens/ones component explicitly; the broader course outcome remains incomplete until its other taught components are checked.

Feedback

For R1 unit confusion, rebuild one ten from named single units. For R2, reverse the exchange while retaining the full collection. For R3, contrast zero extra ones with no tens and no ones. For R4, separate the addition from the regrouping, then pack ten single ones into one ten. Reassess gaps with different keyed applications within the taught range and retain individual evidence privately.

Quantity, name, numeral, and expanded form

- Lesson ID: L-MATH-001-01-03

- Unit: U-MATH-001-01

- Intended learners: 2nd grade

- Status: draft

- Primary art: Arithmetic

- Course outcome focus: 003; review 001 (prefix O-MATH-001-)

- Time estimate: 25 minutes for a core route; optional thousand activity adds a separate 25-minute estimate

- Materials: counters/bundled sticks or base-ten blocks, paper, pencil, place-value chart

- Optional materials: ten established hundred-groups or hundred flats, an enclosing thousand-group, four-column place-value chart

- Authoring review: 2026-10-09; first-unit review

- Readiness review: pending

Prerequisites

For the two-digit route, explain tens/ones and a zero ones position using lesson 1's models. For the three-digit route, also require lesson 2's explanation of hundreds. Choose the corresponding activities below.

Grammar

Distinguish the amount, its spoken name, its numeral, and a sum of place values. Introduce expanded form as another representation of the same amount.

Logic

Test the claim that 608 = 60 + 8 by building both amounts. Find the error in which unit the 6 represents.

Rhetoric

Present one amount in three forms and explain why all three refer to the same quantity.

Activities

1. Retrieval: model 305 and explain its zero position (3 minutes).

2. Model: connect 257, two hundred fifty-seven, and 200 + 50 + 7 (5 minutes).

3. Supported work: convert 340 and 608 between forms (7 minutes).

4. Independent work: write five hundred three as a numeral and expanded form (7 minutes).

5. Exit: present 460 in words and expanded form, then explain the 6 (3 minutes).

Teacher key and next step

- 340: three hundred forty; 300 + 40.

- 608: six hundred eight; 600 + 8 (a zero-tens term is optional).

- Five hundred three: 503; 500 + 3.

- 460: four hundred sixty; 400 + 60; the 6 contributes six tens.

- Accept an accurate spoken explanation even when spelling needs support.

Two-digit route

Use this route when only ones/tens have been taught. It replaces the three-digit examples above; hundreds and the thousand position remain pending.

Stage	Task and mathematical work	Minutes	Teacher key
Retrieval	Model 40 and explain its zero position	3	4 tens, 0 ones
Model	Connect 57, its spoken name, and its place-value sum	5	Fifty-seven; 50 + 7

Stage	Task and mathematical work	Minutes	Teacher key
Supported work	Convert 60 and 68 between forms; test the claim that the 6 in 68 means six single ones	7	Sixty = 60; sixty-eight = 60 + 8; the 6 records six tens
Independent work	Write thirty-nine as a numeral and place-value sum	7	39; 30 + 9
Exit	Name 82, write its expanded form, and explain the 8	3	Eighty-two; 80 + 2; eight tens

Seek agreement among the numeral, spoken name, sum, and model. Accept accurate spoken names without requiring independent spelling. Return to concrete tens/ones if a digit is treated as single ones regardless of its place.

Optional thousand extension

Teach this separate activity only after lesson 2's hundred exchange and the three-digit forms above have been demonstrated. Before starting, ask the learner to build 200, name the units, and explain why one hundred can be unpacked into ten tens. If the unit relationships are uncertain, return to lesson 2. This activity extends outcomes 001 and 003; it is outside the two-digit core route. Allow another 25 minutes or split the work; these are authoring estimates.

Use ten hundred flats or ten established groups, each containing ten tens made from ten ones each. Keep their unit value visible. Use a container or band to enclose those ten groups as one thousand-group, or exchange them for a thousand block with the same value. A token alone is insufficient until its represented amount has been established. Use a chart headed Thousands, Hundreds, Tens, Ones.

Worked examples and concrete exchange

1. **Name and count the groups (3 minutes).** Place nine hundred-groups on the table. Count 100, 200, 300, 400, 500, 600, 700, 800, 900. Add one more hundred-group: ten groups of 100 ones make 1,000 ones. Name the difference between the number of groups (ten) and the number of ones (one thousand).

2. **Model the exchange and the fourth place (5 minutes).** Enclose the ten hundred-groups in one thousand-group. Open it again to check its contents. Explain:

ten hundreds and one thousand both contain 1,000 of the same ones; packaging changes the grouped unit, while the amount remains unchanged. Record the exchange as 10 hundreds = 1 thousand = 1,000 ones. On the chart, place 1 in Thousands and 0 in each lower column. Write 1,000; read "one thousand"; give its place-value sum as 1,000 + 0 + 0 + 0 (or simply 1,000). Compare 900's three positions with 1,000's four: the 1 occupies the new place to the left of Hundreds. The comma separates digits and is not a digit position. The three zeros record no additional hundreds, tens, or ones outside the thousand-group; they do not mean that the amount contains no ones.

3. **Supported grouping and explanation (7 minutes).** Give one set of four hundred-groups and another of six. Ask the learner to combine them, exchange the ten hundreds for one thousand, then unpack it again. Ask: "Did putting them into one group change the amount? How can we check?" Test the claim "Nine hundreds can also be exchanged for one thousand without changing the amount" using nine groups and the required ten. Finish with a teach-back: point to the objects, each chart column, the numeral, name, and sum and explain their agreement to another person.

Independent practice and exit check

Allow 7 minutes for the following practice, without answer cues. Objects, drawings, and spoken explanations are allowed.

Practice	Learner task
T3-I1	Unpack one thousand-group into hundreds. Then unpack every hundred into tens. Record the number of groups at each stage and explain why the amount stays the same.
T3-I2	Hear "one thousand." Fill a blank Thousands/Hundreds/Tens/Ones chart, write its numeral and a sum of place values, and explain each zero.
T3-I3	A person says, "After ten hundreds are packed into one group, there are only 100 ones." Use the contents of the group to test the claim and explain the error.

Exit check (3 minutes): show six hundred-groups beside four hundred-groups. Ask whether together they can fill one thousand-group exactly. Require a model and a preservation reason; then ask which place records the new grouped unit.

Extension teacher key and next step

Stage or item	Teacher key or model evidence
Prerequisite	200 = 2 hundreds; each hundred can be unpacked into 10 tens, each ten into 10 ones
Worked exchange	9 hundreds = 900; adding 1 hundred gives 10 hundreds = 1,000 ones = 1 thousand. Enclosing or exchanging equal-valued groups adds/removes no ones
Supported task	4 hundreds + 6 hundreds = 10 hundreds = 1 thousand. Nine hundreds contain 900 ones, 100 fewer than a thousand, so that exchange would change the amount
T3-I1	1 thousand = 10 hundreds = 100 tens = 1,000 ones. The number of groups changes with the chosen unit; the amount does not
T3-I2	Chart: 1 thousand, 0 hundreds, 0 tens, 0 ones; numeral 1,000; sum 1,000 or 1,000 + 0 + 0 + 0. Four digit positions; zeros mean no additional lower-place groups
T3-I3	False: each of the 10 hundreds contains 100 ones, so the packed group contains 1,000 ones, not 100. The outer packaging does not remove its contents
Exit	6 hundreds + 4 hundreds = 1 thousand exactly; equal-valued packaging preserves 1,000 ones. Thousands is the new position left of Hundreds

Seek correct unit names, equal amounts before and after exchange, a matching four-position representation, and an explanation another person can follow. If any relationship needs review, reopen the concrete group and reconnect the objects and chart before proceeding. Then teach lesson 4's separate count through 1,000. PVX becomes a taught-scope supplement only after both activities and their checks; no learner performance is inferred from their inclusion here.

The prompts are original school-ops drafts. See the reviewed optional practice mapping. Whole returned sets require hundreds; keep this lesson's local two-digit practice and any separate thousand work within their prepared scope. Individual evidence is kept privately; curriculum observations guide the next plan.

Pack implementation and review

The PV2 guide and PV3 guide supplies the route-specific script, independent learner-sheet navigation, complete keys and explicit check/follow-up criteria. The pack index separates learner and teacher print editions. Use only the chosen route's examples; retain this lesson ID. See the 2026-10-27 audit for the record-specific authoring review and remaining readiness decisions. Status remains draft; no learner evidence or delivery schedule is inferred.

Counts, equal steps, and boundaries

- Lesson ID: L-MATH-001-01-04

- Unit: U-MATH-001-01

- Intended learners: 2nd grade

- Status: draft

- Primary art: Arithmetic

- Course outcome focus: 002; review 003 (prefix O-MATH-001-)

- Time estimate: 25 minutes for a core route; optional thousand boundary adds a separate 10-15-minute estimate

- Materials: counters/bundled sticks or base-ten blocks, paper, pencil, place-value chart

- Authoring review: 2026-10-09; first-unit review

- Readiness review: pending

Prerequisites

Read the values used and understand the counted unit. Choose a smaller range when necessary; the repeated-step relationship remains the same.

Grammar

Name starting value, step, sequence, and next value. State explicitly whether the first value is included in the requested number of terms.

Logic

Predict what follows 99 and 199 and justify the regrouping. Compare two sequences with different steps and explain which counted unit changes.

Rhetoric

Teach a counting pattern to another person, saying the rule as well as the values.

Activities

1. Retrieval: count 98 through 102 using objects or a model (3 minutes).

2. Model: start at 230 and count by tens: 230, 240, 250, 260 (5 minutes).

3. Supported work: count by hundreds from 500 to 900; discuss boundaries (7 minutes).

4. Independent work: give five terms by fives starting at 95 (7 minutes).

5. Exit: count 198 through 202; give three terms by tens from 390 and explain the step (3 minutes).

Teacher key and next step

- First count: 98, 99, 100, 101, 102.

- Hundreds: 500, 600, 700, 800, 900.

- Fives: 95, 100, 105, 110, 115, including the starting value.

- Exit: 198, 199, 200, 201, 202; 390, 400, 410.

- Seek an explained repeated step, not only a recited sequence.

- If a boundary causes difficulty, model the exchange and practice a shorter span.

Two-digit route

Use these examples when ones/tens are the taught scope. Include the starting value in every stated number of terms. Hundreds and the thousand boundary are separate later work.

Stage	Task and mathematical work	Minutes	Teacher key
Retrieval	Count by ones from 38 through 42	3	38, 39, 40, 41, 42
Model	Give four terms by tens starting at 20	5	20, 30, 40, 50; one ten added each time
Supported work	Give five terms by fives starting at 25	7	25, 30, 35, 40, 45; step five
Independent work	Give five terms by fives starting at 55	7	55, 60, 65, 70, 75

Stage	Task and mathematical work	Minutes	Teacher key
Exit	Count 58 through 62; give three terms by tens starting at 46 and explain the step	3	58, 59, 60, 61, 62; 46, 56, 66

Ask for the step and a model of the tens boundary, rather than only a recited list.

Optional count through 1,000

Use after lesson 3's thousand activity and check. This adds 10-15 minutes or a separate sitting to the core lesson; it is outside the two-digit and three-digit core routes. Use nine hundred-groups, nine tens, nine ones, one extra one, and a four-column chart. A thousand container or established thousand block represents ten hundreds, as in lesson 3.

1. Model 999 as 9 hundreds + 9 tens + 9 ones. Add one single one and name the new amount. Exchange 10 ones for 1 ten: 9 hundreds + 10 tens + 0 ones. Exchange 10 tens for 1 hundred: 10 hundreds + 0 tens + 0 ones. Finally, exchange 10 hundreds for 1 thousand: 1 thousand + 0 hundreds + 0 tens + 0 ones. Explain why each exchange preserves the amount after the one was added.

2. Supported count: start at 993 and count by ones through 1,000, including both endpoints. At the final step, demonstrate the exchanges on the objects and chart, then explain the same step to another person.

3. Independent practice: count 994 through 1,000 and draw the groups immediately before and after the last step. Ask for the repeated step and all three exchanges. If counting or the exchange needs support, review that part before PVX.

Teacher key: supported count is 993, 994, 995, 996, 997, 998, 999, 1,000; independent count is 994, 995, 996, 997, 998, 999, 1,000. The repeated step adds one one. Before the final step the amount is 999; after adding one it is 1,000. All exchanges then preserve 1,000. Seek the complete sequence, the distinction between adding and regrouping, and the three exchanges with matching digit places. Review lesson 3's group model if the thousand exchange is uncertain. PVX-02 checks this relationship with a different starting point, not further four-digit work.

The prompts are original school-ops drafts. See the reviewed optional practice mapping. Whole returned sets require hundreds; keep this lesson's local two-digit practice and any separate thousand work within their prepared scope. Individual evidence is kept privately; curriculum observations guide the next plan.

Pack implementation and review

The PV2 guide and PV3 guide supplies the route-specific script, independent learner-sheet navigation, complete keys and explicit check/follow-up criteria. The pack index separates learner and teacher print editions. Use only the chosen route's examples; retain this lesson ID. See the 2026-10-27 audit for the record-specific authoring review and remaining readiness decisions. Status remains draft; no learner evidence or delivery schedule is inferred.

Bank workflow for the selected route

Draft • teacher/developer guide • C-MATH-001 / U-MATH-001-01

Use version 1.0.1 after the selected route's prerequisites and local assessment review. Frozen public examples and legacy forms are review artifacts. No generator output is an observed learner result.

From the repository root, with Python 3.11 or later:

```
python -m exercise_bank generate --route PV2 --mode drill --seed pack-practice-01 --duration 90 --output /tmp/pack-pv2-drill

python -m exercise_bank validate /tmp/pack-pv2-drill/manifest.json

python -m exercise_bank generate --route PV2 --mode assessment --seed pack-check-01 --output /tmp/pack-pv2-check

python -m exercise_bank validate /tmp/pack-pv2-check/manifest.json
```

For PV3 use --route PV3 and different new output directories. PVX is never generated. The 90-second drill is an example, not an agreed duration. No real learner identifiers belong in seeds. Existing output directories are not overwritten.

Mode	Placement and administration	Decision
Drill	Optional after instruction in all four sampled relationships, or alongside later review; choose duration locally; print learner sheet before any timer	Review errors and unfinished work; no outcome or speed pass mark
Assessment	An untimed alternate application of A-MATH-001-02 in lesson 6 or reassessment after targeted review; allow extra time or split sittings	Check each outcome group and every model/reason; no aggregate score

Before distribution, inspect all 12 prompts, key/model evidence, range, zero and boundary cases, exposure and accessibility. Validate the manifest. Print only learner.md; keep teacher.md and manifest.json away from learners. The manifest contains answers. A different seed can repeat prompts; inspect freshness against actual prior practice privately. Use A-MATH-001-02 items 01-03 plus the composition in 04 for outcome 001; 04-06 for 002; 07-09 for 003; 10-12 for 004.

Read directions without solution cues. Allow objects, blank chart, speech, drawing, pointing and adult scribing; no calculator. Spelling is not assessed. Recognize

substantive help as support. Give feedback after the attempt. Retain route, generator version, form ID, support and per-outcome judgments privately. A fresh keyed form supports reassessment but has no established psychometric equivalence to another seed.